Credit Risk Portfolio Analytics - Solution Overview

Streamlit dashboard with an LLM 'Ask the Portfolio' assistant and explainable anomaly detection, built on 5 synthetic bank tables (10,000 customers, 14,685 loans).

What the dashboard shows and how an analyst uses it

- **Portfolio** - KPIs (accounts, exposure, default rate, exposure-weighted default rate), model risk segments with actual vs predicted default, default rate by any of 17 dimensions with 95% intervals, **drill-down** to the accounts in a segment, a cross-dimensional heatmap and a vintage trend. Sidebar filters (region, loan type, employment, income, age, segment, year) apply throughout. Use: find where risk and dollars concentrate.
- **Insights** - five computed insights, each with evidence (n shown) and a 'so what'.
- **Account view** - predicted PD, segment, **key drivers** (log-odds contributions that sum to the score), profile, 12-month repayment chart, bureau lines, collections, and any anomaly flag. Use: explain a score to a credit officer.
- **Default model** - AUC/KS/Gini (CV and out-of-time), ROC, calibration by decile, importance (logistic and SHAP), feature-group ablation, PSI monitoring.
- **Anomaly review** and **Ask the Portfolio** - the two AI features (below). **Data quality** - join logic and the register of every issue and decision.

AI feature 1: Ask the Portfolio (LLM)

How it works. One LLM call converts the question into a JSON query plan (or declines it). Python validates the plan against allow-lists and DuckDB computes the figures with bound-parameter SQL. The answer text is built deterministically from the result; optional LLM wording is accepted only if every number matches the table. Output: figures, n, filters, SQL and source. Unanswerable questions (forecasts, causes, city-level, individuals) are declined. No key or an LLM error falls back to a rule-based planner. Only the question, history and category catalogue (and aggregated results) reach the LLM, never customer rows.

Trigger analyst submits a question. **False positive** a valid question declined, or answered with the wrong metric (shown via plan/SQL). **False negative** an unanswerable question answered plausibly; blocked by scope gate, allow-lists and value checks. **Human step** analyst checks plan, SQL and n; marks the answer Correct/Wrong (logged).

AI feature 2: Anomaly detection (explainable, no LLM)

How it works. Four detectors: (A) last 3 months vs the account's own previous 9; (B) top 2% of its peer segment (loan type x income tier); (C) bureau-vs-internal divergence; (D) IsolationForest within each peer segment. Score >= 3 enters the queue with plain-English reasons. 809 accounts (5.5%) are queued; flagged accounts default at 24.8% vs 15.6% (1.59x), though the label is never used.

Trigger batch score >= 3. **False positive** a one-off delay or a legitimately high loan-to-income. **False negative** slow drift that never crosses a threshold, or a thin-file customer with no bureau comparison. **Human step** analyst confirms / dismisses / requests info with a note; append-only audit log; decisions drive threshold recalibration. Nothing is automated.

Key assumptions

- **Exposure** = original loan_amount (no current balance on the loan table). Target *target_default* is taken as given; its horizon is not stated.
- **Bureau is a current snapshot** (62.5% of lines opened after the loan), so this is a behavioural/monitoring score, not an origination score.
- **Collections excluded** from the model (recorded after delinquency; 44.9% dated before origination). customer_since ignored (after the loan for 49.7%).
- Gender and marital status are not model inputs. Synthetic data: results describe this snapshot, not a real bank.

Model results and approach

Model	AUC	KS	Gini	OOT AUC
Logistic (champion)	0.680	0.270	0.360	0.689
LightGBM	0.677	0.259	0.355	0.688

Target target_default (16.1% of 14,685 accounts). **Features** 32 numeric + 3 categorical: origination (income, age, loan size, LTI, rate), bureau (write-offs, DPD, utilisation, thin-file) and 12-month repayment (late/missed counts, streak, DPD, EMI burden). **Validation** 5-fold CV grouped by customer (a customer's loans never straddle train/test; AUC 95% CI 0.669-0.692) plus an out-of-time test (train <= 2021, test > 2021). No class re-weighting, so outputs are real PDs; risk-segment PDs match realised rates. Champion rule: the ML model must beat the baseline by > 0.005 AUC; it did not, so the interpretable logistic model wins. **Honest read:** AUC ~0.68 is modest and near this data's ceiling. Origination data alone gives 0.654; the months 1-6 model still gives 0.673 (no late-window artefact); collections add no measurable AUC (0.6800 -> 0.6800). Use as a ranking/monitoring aid.

Monitoring. Monthly PSI on score and features (<0.10 stable, >0.25 act), AUC/KS on matured cohorts (alert if AUC falls > 0.03 below the CI), predicted-vs-actual by segment, fairness by protected group, anomaly confirm/dismiss rates; retrain via champion-challenger.

Top 3 portfolio insights

1. Mortgages dominate dollar risk. 9.8% of accounts but 52% of exposure; default 24.7% vs 16.1%; 65% of defaulted exposure. The value-weighted default rate is 21.1% vs 16.1% by count. *So what:* govern mortgage underwriting and early arrears first; report both rates.

2. Unemployment compounds low income (cross-cut employment x income). Unemployed default at 35% (n=718). Below 40k income: unemployed 41.7% vs salaried 23.9%; income <40k overall 32.4% vs 9.5% at 100k+. *So what:* an income-only cut-off misses employment risk; tighten terms for unemployed applicants.

3. Delinquency is an early-warning ladder. Default rises from 8.1% (no late/missed month) to 28.7% (4+; 13.9% of accounts), visible by month 6. *So what:* start collections outreach at the second delinquent month.

To go to production, we would need from the bank's IT and data teams

- **Point-in-time data:** as-of-date bureau pulls, loan snapshots, a defined default definition and horizon, and event timestamps that pass integrity checks.
- **Access and platform:** read-only feeds from the core banking, bureau and collections systems on a daily/monthly schedule; a governed data lake; SSO and role-based access; approved hosting (private VPC) and an enterprise LLM endpoint (e.g. Azure OpenAI) with logging and no customer rows sent.
- **Governance:** model-risk validation, fair-lending review of age, a database for review decisions and audit logs, outcome labels fed back from collections, monitoring owners and thresholds.

Data quality: 13 issues found in 55 checks; all handled and logged in the dashboard (Data quality tab).